

# Deploying Metabase on AWS

by Mohamad Ikhsan Nurulloh



Metabase



# Disclaimer

- **No Endorsement:** This guide is for educational purposes only and does not imply endorsement by Metabase or AWS.
- **Testing Only:** Intended for testing and development, not production environments.
- **Refer to Official Docs:** Always consult official Metabase and AWS documentation for production use.
- **Proceed at Your Own Risk:** Use this guide at your own risk and be aware of potential deployment implications.

# What is Metabase?



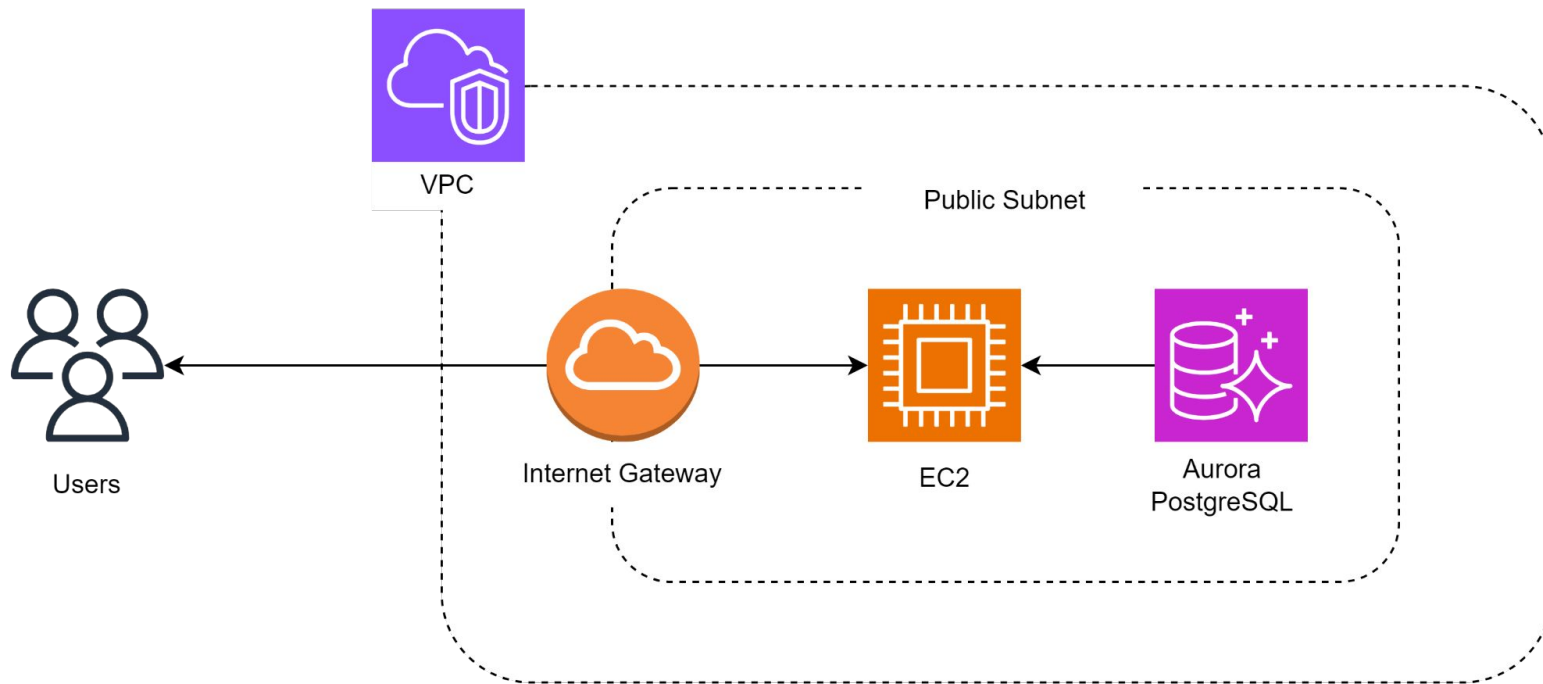
**Metabase** is an open-source BI tool for visualizing data through charts and dashboards. It offers a Query Builder, customizable dashboards, and alerts. Ideal for business reporting, quick analysis, and data exploration, Metabase simplifies data-driven decision-making.

# Why Metabase

- **User-Friendly:** Metabase's intuitive interface lets users create and share dashboards without extensive technical knowledge.
- **Open Source:** Free and customizable, Metabase benefits from a large, active community.
- **Powerful Analytics:** Offers robust analytics, complex queries, and diverse data visualizations.
- **Seamless Integration:** Integrates smoothly with various databases and data sources.
- **Collaboration:** Promotes teamwork by enabling easy sharing of dashboards and reports.

# Metabase Architecture

# Metabase Architecture



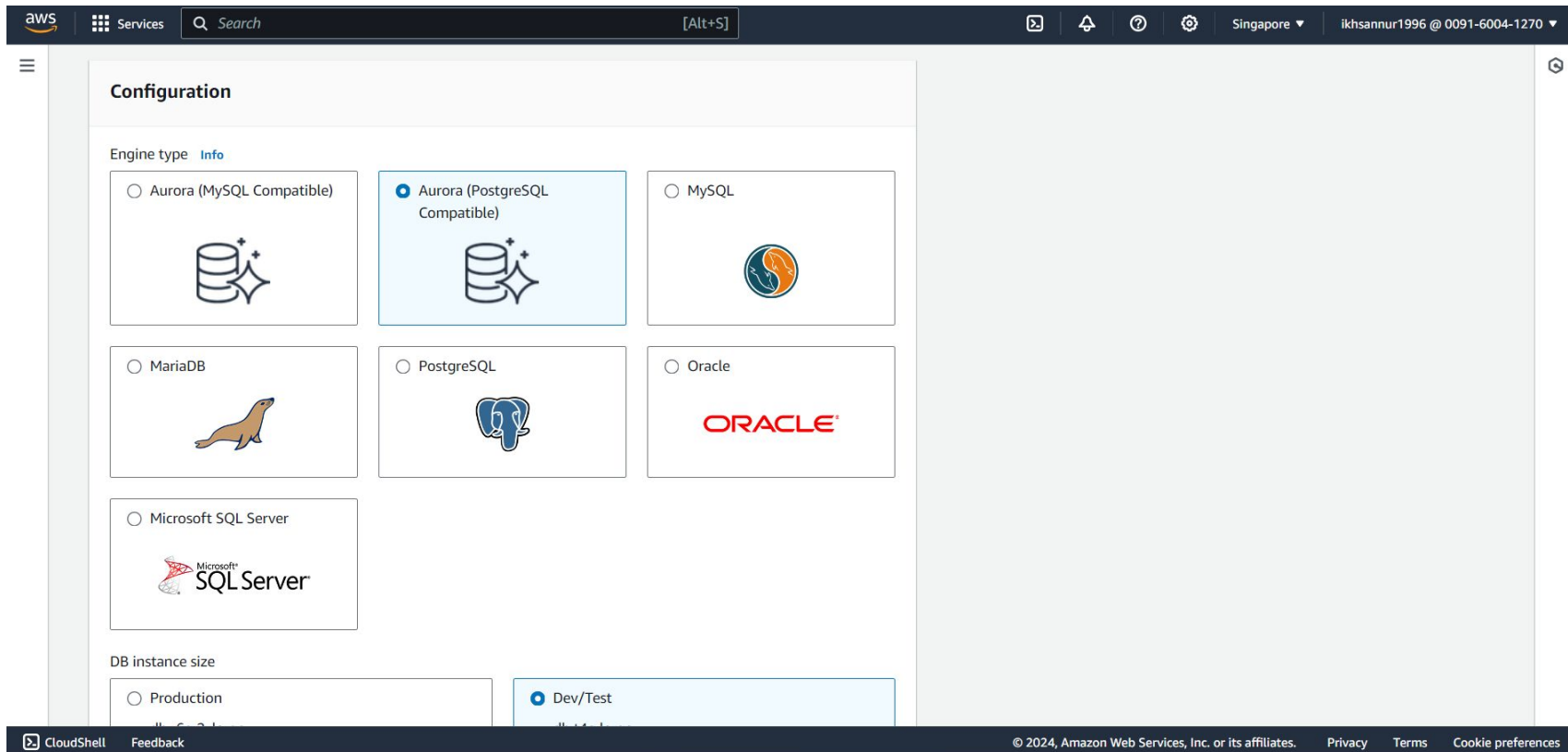
**Overall Architecture:** Users access the Metabase application hosted on an EC2 instance in a public subnet. The EC2 instance communicates with the Aurora PostgreSQL database within the VPC to retrieve and analyze data. The internet gateway allows the EC2 instance to be accessible from the internet, enabling users to interact with Metabase from anywhere.

# Details Metabase Architecture

- **Users:** End-users (data analysts, business users) who interact with Metabase to query data, generate reports, and visualize insights.
- **VPC (Virtual Private Cloud):** A secure and scalable AWS environment for deploying resources like EC2 and RDS, ensuring network isolation and secure communication.
- **Internet Gateway:** Enables EC2 instances in the VPC to connect to the internet, allowing users to access Metabase from anywhere.
- **EC2 (Elastic Compute Cloud):** Hosts the Metabase application, processing user requests, querying databases, and serving the interface.
- **Public Subnet:** A subnet that allows the EC2 instance running Metabase to be accessible from the internet.
- **Aurora PostgreSQL:** A managed PostgreSQL database engine that stores and supports data queries and analysis in Metabase.

# Setting Up RDS Aurora (PostgreSQL Compatible)

# Creating RDS - Aurora (PostgreSQL Compatible)



The screenshot displays the AWS Management Console interface for configuring a new Amazon RDS instance. The top navigation bar includes the AWS logo, a 'Services' menu, a search bar, and the user's account information (ikhsannur1996 @ 0091-6004-1270) in the Singapore region. The main content area is titled 'Configuration' and features a grid of database engine options under the 'Engine type' section. The 'Aurora (PostgreSQL Compatible)' option is selected, indicated by a blue border and a blue radio button. Other visible options include Aurora (MySQL Compatible), MySQL, MariaDB, PostgreSQL, Oracle, and Microsoft SQL Server. Below the engine type grid, the 'DB instance size' section shows 'Dev/Test' as the selected instance class. The bottom of the console shows a footer with 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc.

**Configuration**

Engine type [Info](#)

- ☐ Aurora (MySQL Compatible)
- ☒ Aurora (PostgreSQL Compatible)
- ☐ MySQL
- ☐ MariaDB
- ☐ PostgreSQL
- ☐ Oracle
- ☐ Microsoft SQL Server

DB instance size

- ☐ Production
- ☒ Dev/Test

Detailed steps for creating an Amazon RDS instance using Aurora PostgreSQL, including selecting the engine type and configuring the database instance.

# Select DB Instance Size

aws

Services

Search

[Alt+S]

📧 🔔 ⓘ ⚙️

Singapore ▼

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☰

DB instance size

☐ Production  
db.r6g.2xlarge  
8 vCPUs  
64 GiB RAM  
1.253 USD/hour

☒ Dev/Test  
db.t4g.large  
2 vCPUs  
8 GiB RAM  
0.229 USD/hour

DB cluster identifier

Enter a name for your DB cluster. The name must be unique across all DB clusters owned by your AWS account in the current AWS Region.

mydatabase

The DB cluster identifier is case-insensitive, but is stored as all lowercase (as in "mydbcluster"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Master username [Info](#)

Type a login ID for the master user of your DB instance.

postgres

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*  
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed  
Create your own password or have RDS create a password that you manage.

☐ Auto generate password  
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

\*\*\*\*\*

Password strength Weak

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' \* @

Confirm master password [Info](#)

CloudShell

Feedback

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This page covers the configuration options for the DB instance, including choosing the instance size (e.g., production or dev/test), specifying the DB cluster identifier, and setting the master username.

# Credentials Management

The screenshot shows the 'Credentials management' page in the AWS console. The page title is 'Credentials management'. Below the title, there is a sub-header: 'You can use AWS Secrets Manager or manage your master user credentials.' There are two radio buttons for selecting the management method: 'Managed in AWS Secrets Manager - most secure' (which is selected) and 'Self managed'. Below these, there is a checkbox for 'Auto generate password'. The 'Master password' field is visible with a strength indicator showing 'Weak'. There is also a 'Confirm master password' field. Below the password fields, there are two expandable sections: 'Set up EC2 connection - optional' and 'View default settings for Easy create'. At the bottom, there is a blue information box stating: 'You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.' At the very bottom, there are 'Cancel' and 'Create database' buttons.

aws Services Search [Alt+S]

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*  
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed  
Create your own password or have RDS create a password that you manage.

☐ Auto generate password  
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength **Weak**

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / \* @

Confirm master password [Info](#)

► **Set up EC2 connection - optional**  
You can also set up a connection to an EC2 instance after creating the database. Go to the database list page or the database details page, choose **Actions**, and then choose **Set up to EC2 connection**.

► **View default settings for Easy create**  
Easy create sets the following configurations to their default values, some of which can be changed later. If you want to change any of these settings now, use [Standard create](#).

**i** You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

Cancel **Create database**

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Options for managing master user credentials, including using AWS Secrets Manager for secure password management or self-managing the credentials.

# RDS Aurora PostgreSQL Success

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Services

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Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

Recommendations

Certificate update

RDS > Databases > mydatabase > mydatabase-instance-1

mydatabase-instance-1

Modify

Actions

Related

Filter by databases

| DB identifier         | Status    | Role             | Engine            | Region & AZ     | Size         | Recommendation |
|-----------------------|-----------|------------------|-------------------|-----------------|--------------|----------------|
| mydatabase            | Available | Regional cluster | Aurora PostgreSQL | ap-southeast-1  | 1 instance   |                |
| mydatabase-instance-1 | Available | Writer instance  | Aurora PostgreSQL | ap-southeast-1c | db.t4g.large |                |

Connectivity & security

Monitoring

Logs & events

Configuration

Maintenance & backups

Tags

Recommendations

Connectivity & security

Endpoint & port

Endpoint

mydatabase-instance-1.cf66iaek0a7y.ap-southeast-1.rds.amazonaws.com

Port

5432

Networking

Availability Zone

ap-southeast-1c

VPC

vpc-093bba71c48d18651

Subnet group

default-vpc-093bba71c48d18651

Security

VPC security groups

default (sg-0aefb0b67b83dcdc4)

Active

Publicly accessible

No

Certificate authority

rds-ca-rsa2048-g1

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Confirmation of the successful deployment of the RDS Aurora PostgreSQL instance, with details about the database instance, such as endpoint, port, and availability zone.

# Enable Public Access

aws

Services

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☒ IPv4

Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode

Your resources can communicate over IPv4, IPv6, or both.

DB subnet group

default-vpc-093bba71c48d18651

Security group

List of DB security groups to associate with this DB instance.

Choose security groups

default

Certificate authority

Info

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)

Expiry: May 22, 2061

Additional configuration

Public access

☒ Publicly accessible

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ Not publicly accessible

No IP address is assigned to the DB instance. EC2 instances and devices outside the VPC can't connect.

Database port

Specify the TCP/IP port that the DB instance will use for application connections. The application connection string must specify the port number. The DB security group and your firewall must allow connections to the port. [Learn more](#)

5432

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Enabling public access to the database and configuring security groups.

# Setting Up Amazon EC2

# Launch EC2 Instance

**Launch an instance** [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

**Name and tags** [Info](#)

Name  
metabase [Add additional tags](#)

**▼ Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

**Recents** | **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Li [Browse more AMIs](#)

**▼ Summary**

Number of instances [Info](#)  
1

**Software Image (AMI)**  
Amazon Linux 2023.5.2...[read more](#)  
ami-0a6b545f62129c495

**Virtual server type (instance type)**  
t2.micro

**Firewall (security group)**  
New security group

**Storage (volumes)**  
1 volume(s) - 8 GiB

Cancel **Launch instance** [Review commands](#)

Selecting the Amazon Machine Image (AMI), instance type, and configuring security group settings.

# Choose Amazon Machine Image

aws

Services

Search

[Alt+S]

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▼ Application and OS Images (Amazon Machine Image) Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Q Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Li

SUSE

Q

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI

ami-0a6b545f62129c495 (64-bit (x86), uefi-preferred) / ami-080f155ef23f03e33 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

▼ Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.5.2...read more

ami-0a6b545f62129c495

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Review commands

CloudShell

Feedback

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Selecting the Amazon Machine Image (AMI), instance type, and configuring security group settings.

# Select Instance Type

## ▼ Instance type [Info](#) | [Get advice](#)

### Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0146 USD per Hour

On-Demand Windows base pricing: 0.0192 USD per Hour

On-Demand RHEL base pricing: 0.029 USD per Hour

On-Demand SUSE base pricing: 0.0146 USD per Hour

☒ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

Selecting the Amazon Machine Image (AMI), instance type, and configuring security group settings.

# Key Pair Configuration

## ▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended)

Default value ▼



[Create new key pair](#)

Options for managing AWS Key Pair for secure login management.

# EC2 Network Configuration

Configuration of security group rules to allow access to the Metabase port, including setting up rules for HTTP, HTTPS, and SSH traffic.

▼ Network settings [Info](#)

Edit

Network [Info](#)

vpc-093bba71c48d18651

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere  
0.0.0.0/0

☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

# Storage Configuration

aws

Services

Search

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Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

▼ Configure storage

Info

Advanced

1x

8

GiB

gp3

Root volume (Not encrypted)

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Edit

► Advanced details

Info

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.5.2...read more

ami-0a6b545f62129c495

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Review commands

CloudShell

Feedback

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Options for Configuring Storage Size.

# EC2 Instance Success Running

The screenshot displays the AWS Management Console interface for an EC2 instance. The left sidebar shows navigation options like EC2 Dashboard, Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, Lifecycle Manager, Network & Security, Security Groups, Elastic IPs, Placement Groups, and Key Pairs. The main content area shows the 'Instance summary for i-03204c0ca8fe2f406 (metabase)' with a status of 'Running'. The summary includes details such as Instance ID, Public IPv4 address (175.41.178.121), Private IPv4 addresses (172.31.37.128), Instance state (Running), Public IPv4 DNS (ec2-175-41-178-121.ap-southeast-1.compute.amazonaws.com), Private IP DNS name (ip-172-31-37-128.ap-southeast-1.compute.internal), Hostname type (IP name: ip-172-31-37-128.ap-southeast-1.compute.internal), Answer private resource DNS name (IPv4 (A)), Auto-assigned IP address (175.41.178.121 [Public IP]), IAM Role, IMDSv2 Required, Instance type (t2.micro), VPC ID (vpc-093bba71c48d18651), Subnet ID (subnet-0fabf15788bfbfd73f), and Instance ARN (arn:aws:ec2:ap-southeast-1:009160041270:instance/i-03204c0ca8fe2f406). The bottom of the console shows the CloudShell and Feedback buttons, and the footer indicates the copyright for Amazon Web Services, Inc. or its affiliates, along with links for Privacy, Terms, and Cookie preferences.

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EC2 Dashboard X

EC2 Global View

Events

▼ Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations

▼ Images

- AMIs
- AMI Catalog

▼ Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs

EC2 > Instances > i-03204c0ca8fe2f406

**Instance summary for i-03204c0ca8fe2f406 (metabase)** Info

Updated less than a minute ago

Connect Instance state Actions

|                                                                            |                                                                                      |                                                                                                    |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Instance ID<br>i-03204c0ca8fe2f406 (metabase)                              | Public IPv4 address<br>175.41.178.121   open address                                 | Private IPv4 addresses<br>172.31.37.128                                                            |
| IPv6 address<br>-                                                          | Instance state<br>Running                                                            | Public IPv4 DNS<br>ec2-175-41-178-121.ap-southeast-1.compute.amazonaws.com   open address          |
| Hostname type<br>IP name: ip-172-31-37-128.ap-southeast-1.compute.internal | Private IP DNS name (IPv4 only)<br>ip-172-31-37-128.ap-southeast-1.compute.internal  | Elastic IP addresses<br>-                                                                          |
| Answer private resource DNS name<br>IPv4 (A)                               | Instance type<br>t2.micro                                                            | AWS Compute Optimizer finding<br>Opt-in to AWS Compute Optimizer for recommendations.   Learn more |
| Auto-assigned IP address<br>175.41.178.121 [Public IP]                     | VPC ID<br>vpc-093bba71c48d18651                                                      | Auto Scaling Group name<br>-                                                                       |
| IAM Role<br>-                                                              | Subnet ID<br>subnet-0fabf15788bfbfd73f                                               |                                                                                                    |
| IMDSv2<br>Required                                                         | Instance ARN<br>arn:aws:ec2:ap-southeast-1:009160041270:instance/i-03204c0ca8fe2f406 |                                                                                                    |

Details Status and alarms Monitoring Security Networking Storage Tags

CloudShell Feedback

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Confirmation of the successful launch of the EC2 instance, with details about the instance, such as instance ID, public and private IP addresses, and instance state.

# Custom Inbound Rules for Metabase Port

The screenshot displays the AWS Management Console interface for a security group named 'sg-053a5c2740783d442 - launch-wizard-1'. The left sidebar shows the navigation menu with categories like EC2 Dashboard, Instances, Images, Elastic Block Store, and Network & Security. The main content area is divided into a 'Details' section and a 'Rules' section.

**Details Section:**

- Security group name:** launch-wizard-1
- Security group ID:** sg-053a5c2740783d442
- Description:** launch-wizard-1 created 2024-08-14T12:00:18.750Z
- VPC ID:** vpc-093bba71c48d18651
- Owner:** 009160041270
- Inbound rules count:** 4 Permission entries
- Outbound rules count:** 1 Permission entry

**Rules Section:**

The 'Inbound rules' tab is active, showing a table with 4 rules. The table has columns for Name, Security group rule..., IP version, Type, Protocol, and Port range.

| Name | Security group rule... | IP version | Type       | Protocol | Port range |
|------|------------------------|------------|------------|----------|------------|
| -    | sgr-0aff923425986d2a2  | IPv4       | HTTP       | TCP      | 80         |
| -    | sgr-0ea0f8bbb34ea16d2  | IPv4       | Custom TCP | TCP      | 3000       |
| -    | sgr-03432b26f12a7bd... | IPv4       | HTTPS      | TCP      | 443        |
| -    | sgr-01b99c25242a461f7  | IPv4       | SSH        | TCP      | 22         |

Configuration of security group rules to allow access to the Metabase port, including setting up rules for HTTP, HTTPS, and SSH traffic.

# Connect to Instance

The screenshot shows the AWS Management Console interface for connecting to an EC2 instance. The top navigation bar includes the AWS logo, 'Services', a search bar, and user information. The main heading is 'Connect to instance' with an 'Info' link. Below this, a sub-header states: 'Connect to your instance i-03204c0ca8fe2f406 (metabase) using any of these options'. There are four tabs: 'EC2 Instance Connect' (selected), 'Session Manager', 'SSH client', and 'EC2 serial console'. A yellow warning box indicates that Port 22 (SSH) is open to all IPv4 addresses and provides instructions for restricting access. The instance details section shows the Instance ID 'i-03204c0ca8fe2f406 (metabase)'. Under 'Connection Type', the 'Connect using EC2 Instance Connect' option is selected. The 'Public IP address' is listed as '175.41.178.121'. The 'Username' field contains 'ec2-user'. A blue note box states that 'ec2-user' is the default username. At the bottom, there are 'Cancel' and 'Connect' buttons.

aws Services [Alt+S] Singapore ikhsannur1996 @ 0091-6004-1270

## Connect to instance [Info](#)

Connect to your instance i-03204c0ca8fe2f406 (metabase) using any of these options

[EC2 Instance Connect](#) Session Manager SSH client EC2 serial console

**⚠ Port 22 (SSH) is open to all IPv4 addresses**

Port 22 (SSH) is currently open to all IPv4 addresses, indicated by **0.0.0.0/0** in the inbound rule in [your security group](#). For increased security, consider restricting access to only the EC2 Instance Connect service IP addresses for your Region: 3.0.5.32/29. [Learn more](#).

Instance ID

i-03204c0ca8fe2f406 (metabase)

Connection Type

☒ **Connect using EC2 Instance Connect**

Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.

☐ **Connect using EC2 Instance Connect Endpoint**

Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IP address

175.41.178.121

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

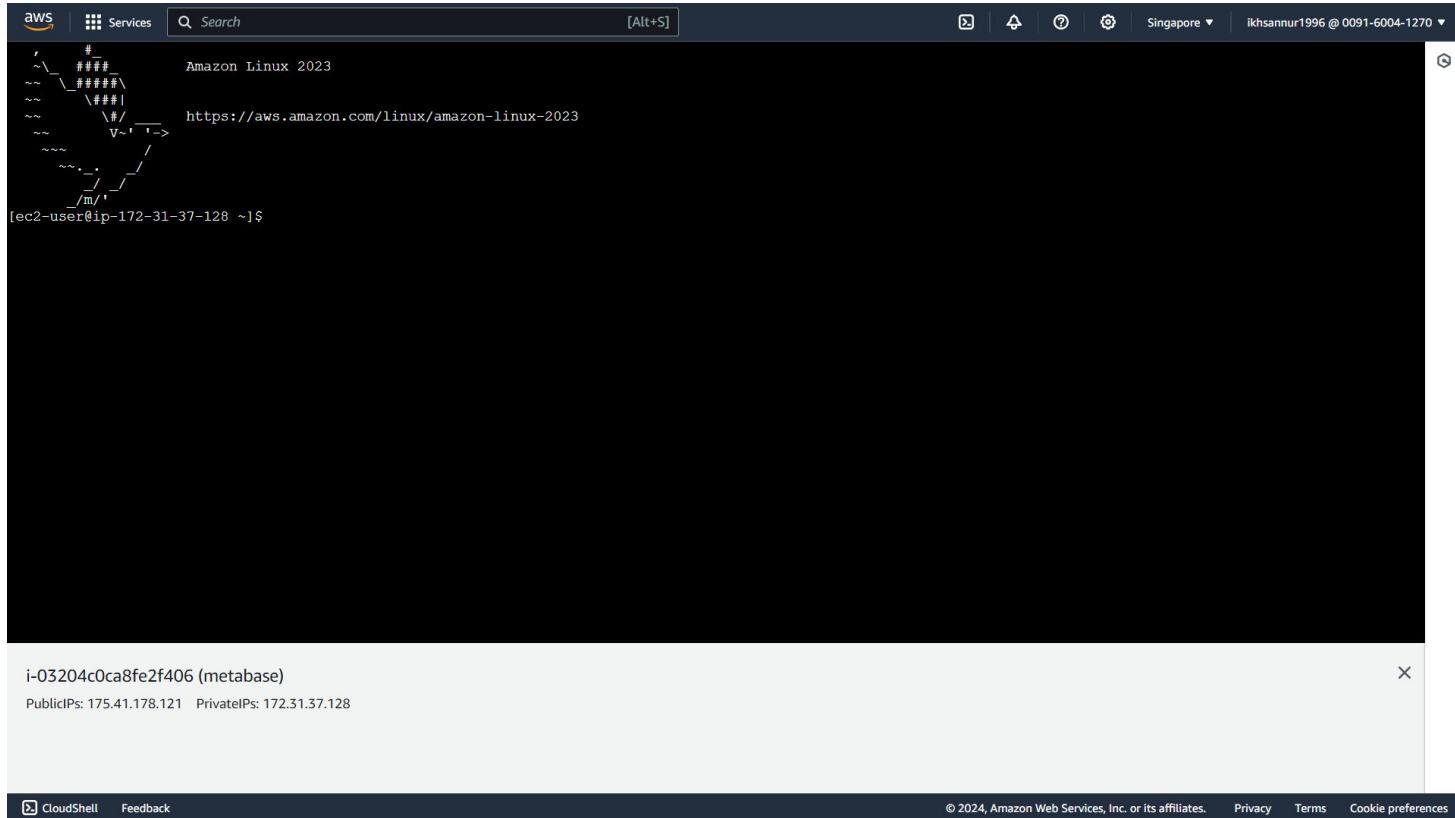
**Note:** In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel **Connect**

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Instructions for connecting to the EC2 instance using various methods, such as EC2 Instance Connect, Session Manager, SSH client, and EC2 serial console.

# SSH Opened



# Setting Up Metabase on Amazon EC2

METABASE OPEN SOURCE

# Run your own free self-hosted instance of Metabase



## Docker Image

Metabase provides an official Docker image via Dockerhub that can be used for deployments on any system that is running Docker. Here's a one-liner that will start a container running Metabase.

```
docker run -d -p 3000:3000 --name metabase metabase/metabase
```

[Running on Docker documentation >](#)



## JAR file

1. Download Metabase.
2. Create a new directory and move the Metabase JAR into it.
3. Change into your new Metabase directory and run the JAR.

```
java -jar metabase.jar
```

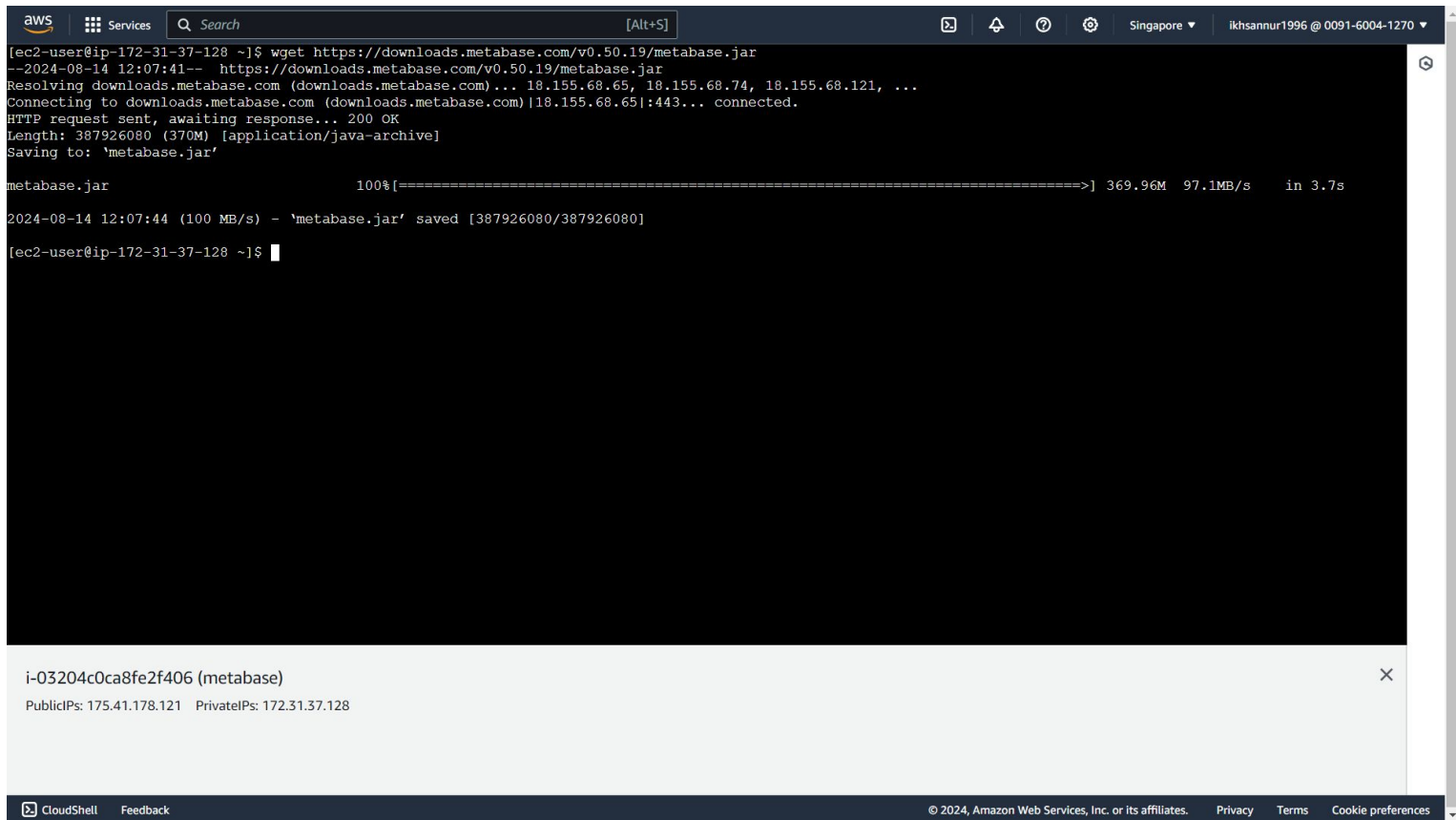
Metabase will log its progress in the terminal as it starts up. Wait until you see "Metabase Initialization Complete" and visit localhost:3000.

[Running the Jar File documentation >](#)

Steps to install Metabase on the EC2 instance, including downloading the Metabase application and setting up the necessary configurations. If Java isn't installed, you'll need to install Java before you can run Metabase.

<https://www.metabase.com/start/oss/>

# Downloading Metabase



```
aws Services Search [Alt+S] Singapore ikhsannur1996 @ 0091-6004-1270
[ec2-user@ip-172-31-37-128 ~]$ wget https://downloads.metabase.com/v0.50.19/metabase.jar
--2024-08-14 12:07:41-- https://downloads.metabase.com/v0.50.19/metabase.jar
Resolving downloads.metabase.com (downloads.metabase.com)... 18.155.68.65, 18.155.68.74, 18.155.68.121, ...
Connecting to downloads.metabase.com (downloads.metabase.com)|18.155.68.65|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 387926080 (370M) [application/java-archive]
Saving to: 'metabase.jar'

metabase.jar          100%[=====>] 369.96M  97.1MB/s   in 3.7s

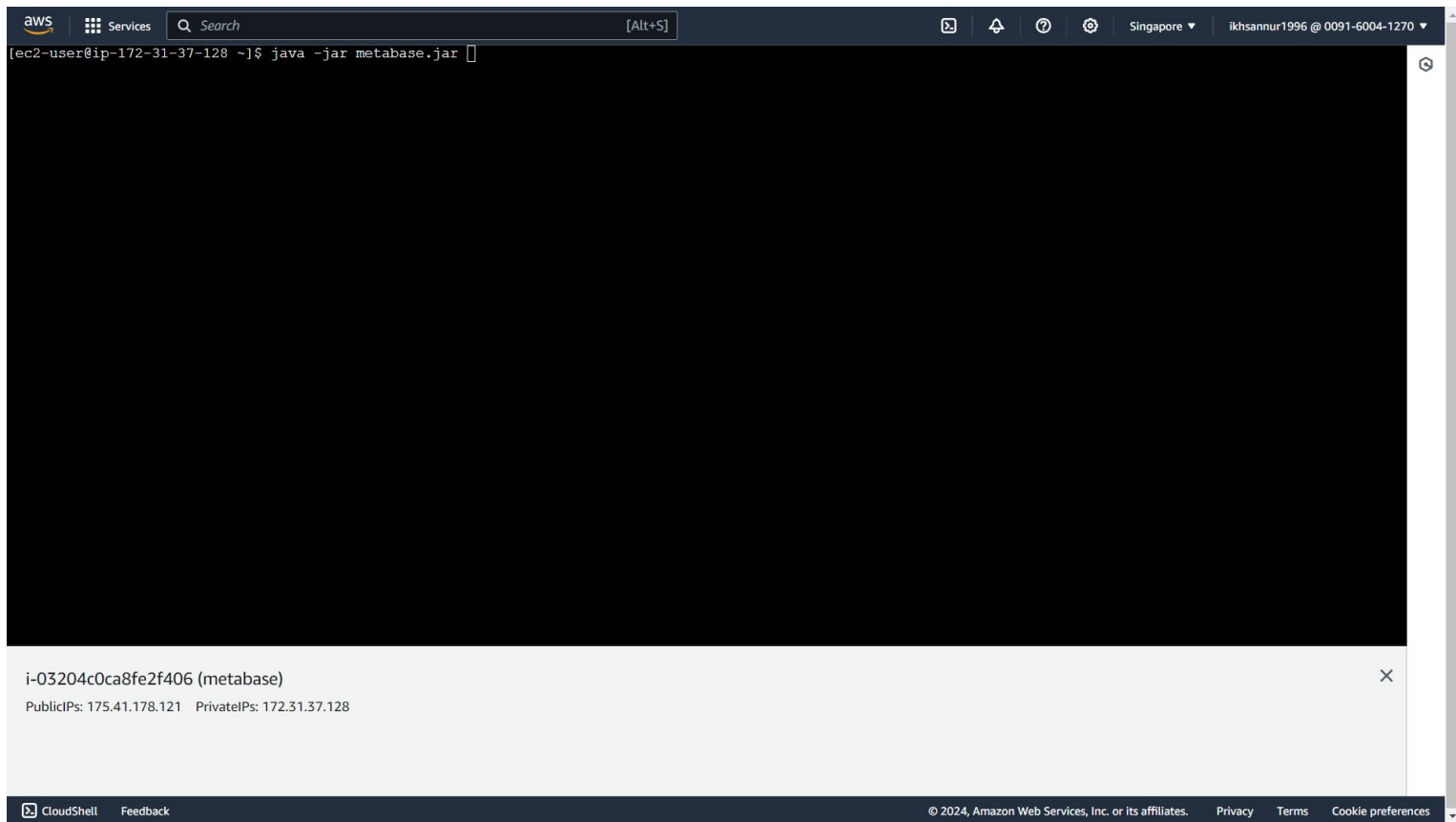
2024-08-14 12:07:44 (100 MB/s) - 'metabase.jar' saved [387926080/387926080]

[ec2-user@ip-172-31-37-128 ~]$
```

i-03204c0ca8fe2f406 (metabase)  
PublicIPs: 175.41.178.121 PrivateIPs: 172.31.37.128

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# Run The Metabase



# Metabase Success Running

```
aws Services Search [Alt+S] Singapore ikhsannur1996 @ 0091-6004-1270
2024-08-14 12:12:53,724 INFO jdbcjobstore.JobStoreTX :: JobStoreTX initialized.
2024-08-14 12:12:53,724 INFO core.QuartzScheduler :: Scheduler meta-data: Quartz Scheduler (v2.3.2) 'MetabaseScheduler' with instanceId 'ip-172-31-37-128.ap-south-east-1.compute.internal1723637573699'
Scheduler class: 'org.quartz.core.QuartzScheduler' - running locally.
NOT STARTED.
Currently in standby mode.
Number of jobs executed: 0
Using thread pool 'org.quartz.simpl.SimpleThreadPool' - with 10 threads.
Using job-store 'org.quartz.impl.jdbcjobstore.JobStoreTX' - which supports persistence. and is clustered.

2024-08-14 12:12:53,725 INFO impl.StdSchedulerFactory :: Quartz scheduler 'MetabaseScheduler' initialized from default resource file in Quartz package: 'quartz.properties'
2024-08-14 12:12:53,725 INFO impl.StdSchedulerFactory :: Quartz scheduler version: 2.3.2
2024-08-14 12:12:53,914 INFO core.QuartzScheduler :: Scheduler MetabaseScheduler_$_ip-172-31-37-128.ap-south-east-1.compute.internal1723637573699 paused.
2024-08-14 12:12:53,916 INFO metabase.task :: Task scheduler initialized into standby mode.
2024-08-14 12:12:53,920 INFO metabase.task :: Initializing task Cache
2024-08-14 12:12:53,923 INFO metabase.task :: Initializing task SyncDatabases
2024-08-14 12:12:53,969 INFO task.sync-databases :: Updated default schedules for 0 databases
2024-08-14 12:12:53,969 INFO metabase.task :: Initializing task PersistRefresh
2024-08-14 12:12:54,018 INFO metabase.task :: Initializing task CheckForNewVersions
2024-08-14 12:12:54,031 INFO metabase.task :: Initializing task PersistPrune
2024-08-14 12:12:54,032 INFO metabase.task :: Initializing task SendAnonymousUsageStats
2024-08-14 12:12:54,038 INFO metabase.task :: Initializing task ModelIndexValues
2024-08-14 12:12:54,039 INFO metabase.task :: Initializing task RefreshSlackChannelsAndUsers
2024-08-14 12:12:54,051 INFO metabase.task :: Initializing task TruncateAuditTables
2024-08-14 12:12:54,054 INFO metabase.task :: Initializing task SendPulses
2024-08-14 12:12:54,064 INFO metabase.task :: Initializing task SendFollowUpEmails
2024-08-14 12:12:54,066 INFO metabase.task :: Initializing task SendCreatorSentimentEmails
2024-08-14 12:12:54,080 INFO metabase.task :: Initializing task SendLegacyNoSelfServiceEmail
2024-08-14 12:12:54,081 INFO metabase.task :: Initializing task TaskHistoryCleanup
2024-08-14 12:12:54,083 INFO metabase.task :: Initializing task SendWarnPulseRemovalEmail
2024-08-14 12:12:54,094 INFO core.QuartzScheduler :: Scheduler MetabaseScheduler_$_ip-172-31-37-128.ap-south-east-1.compute.internal1723637573699 started.
2024-08-14 12:12:54,101 INFO metabase.task :: Task scheduler started
2024-08-14 12:12:54,101 INFO metabase.core :: Metabase Initialization COMPLETE in 1.0 mins
2024-08-14 12:12:54,137 INFO task.refresh-slack-channel-user-cache :: Slack is not configured, not refreshing slack user/channel cache.
```

i-03204c0ca8fe2f406 (metabase)

PublicIPs: 175.41.178.121 PrivateIPs: 172.31.37.128

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Logs and details of the Metabase initialization process, including the setup of the Quartz scheduler and various Metabase tasks. Metabase will run in port 3000.

# Metabase Initialisation and Configuration

# Welcome to Metabase



## Welcome to Metabase

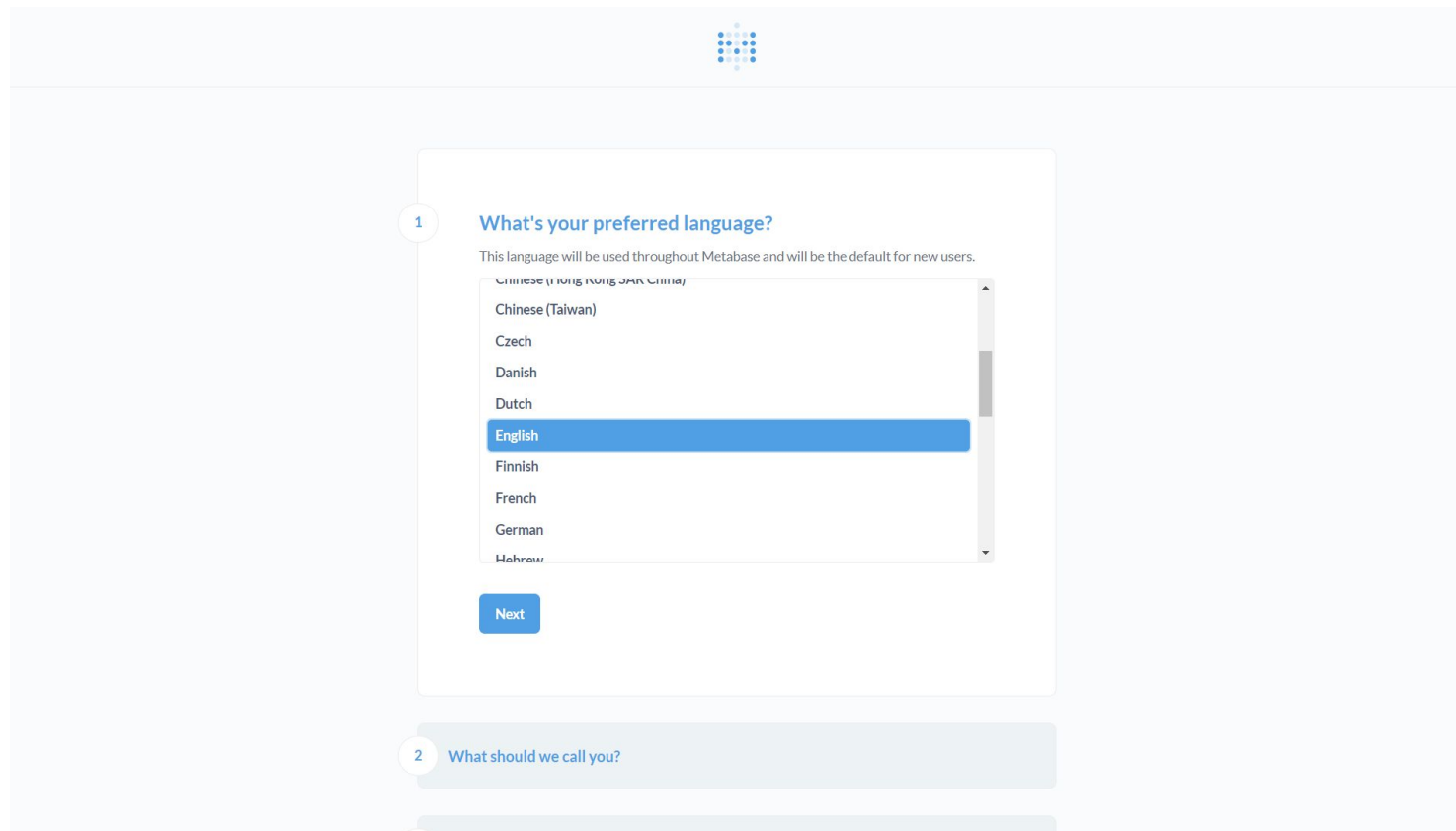
Looks like everything is working. Now let's get to know you, connect to your data, and start finding you some answers!

Let's get started

If you feel stuck, our [getting started guide](#) is just a click away.

Initial setup steps for Metabase, including selecting the preferred language, entering user information, and setting up a password.

# Language Configuration



The image shows a screenshot of the Metabase initial setup interface. At the top center is the Metabase logo, a 4x4 grid of blue dots. Below it is a large white card with rounded corners. On the left side of the card is a small circle containing the number '1'. To the right of this circle is the heading 'What's your preferred language?' in blue. Below the heading is a paragraph: 'This language will be used throughout Metabase and will be the default for new users.' Underneath this text is a scrollable list of languages. The visible languages are: Chinese (Hong Kong SAR China), Chinese (Taiwan), Czech, Danish, Dutch, English (highlighted with a blue background), Finnish, French, German, and Hebrew. At the bottom of the language list is a blue button labeled 'Next'. Below the white card is a light blue bar containing a circle with the number '2' and the text 'What should we call you?'. Below this bar is another light blue bar, which is currently empty.

1 What's your preferred language?

This language will be used throughout Metabase and will be the default for new users.

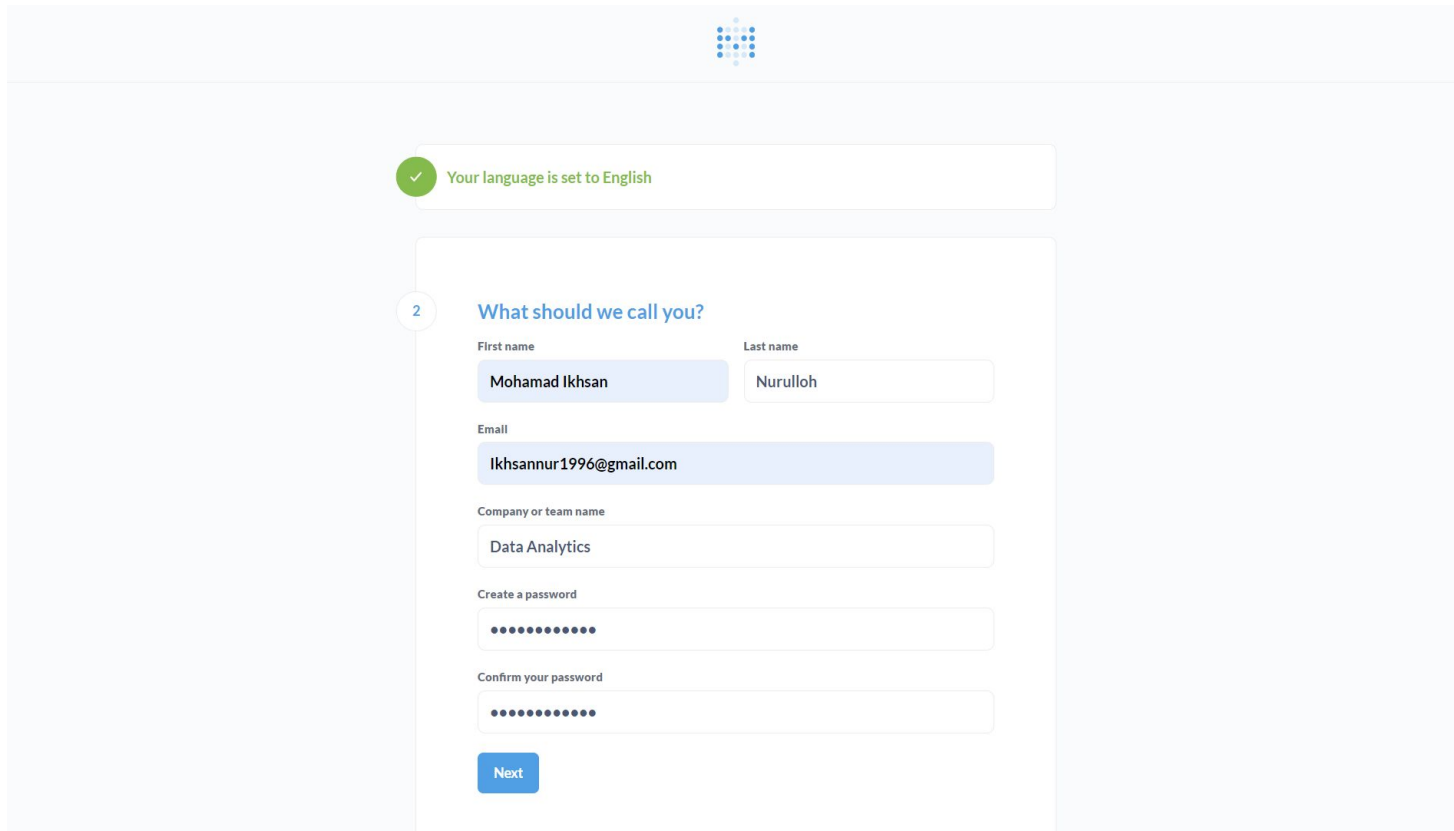
- Chinese (Hong Kong SAR China)
- Chinese (Taiwan)
- Czech
- Danish
- Dutch
- English
- Finnish
- French
- German
- Hebrew

Next

2 What should we call you?

Initial setup steps for Metabase, including selecting the preferred language, entering user information, and setting up a password.

# User Information and Setting Up a Password



The image shows the initial setup screen for Metabase. At the top center is the Metabase logo, a 4x4 grid of blue dots. Below it is a green notification box with a checkmark icon and the text "Your language is set to English". The main content area is a white card with a blue border. On the left side of the card is a blue circle with the number "2". The title of the card is "What should we call you?". The form contains several input fields: "First name" with the value "Mohamad Ikhsan", "Last name" with the value "Nurulloh", "Email" with the value "Ikhsannur1996@gmail.com", "Company or team name" with the value "Data Analytics", "Create a password" with a masked password of 10 dots, and "Confirm your password" with a masked password of 10 dots. At the bottom of the card is a blue "Next" button.

✓ Your language is set to English

2 What should we call you?

First name Last name

Mohamad Ikhsan Nurulloh

Email

Ikhsannur1996@gmail.com

Company or team name

Data Analytics

Create a password

Confirm your password

Next

Initial setup steps for Metabase, including selecting the preferred language, entering user information, and setting up a password.

# What will you use Metabase for?



✓ Your language is set to English

✓ Hi, Mohamad Ikhsan . Nice to meet you!

3 **What will you use Metabase for?**  
Let us know your plans with Metabase so that we can best guide you

- ☒ Self-service analytics for my own company
- ☐ Embedding analytics into my application
- ☐ A bit of both
- ☐ Not sure yet

Next

4 Add your data

# Add Data Connection

4 **Add your data**

Are you ready to start exploring your data? Add it below.  
Not ready? Skip and play around with our Sample Database.

Search for a database...

|                                                                                                      |                                                                                                 |                                                                                                   |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <br>MySQL           | <br>PostgreSQL | <br>SQL Server |
| <br>Amazon Redshift | <br>BigQuery   | <br>Snowflake  |

Show more options ▾

[I'll add my data later](#)

5 **Usage data preferences**

If you feel stuck, our [getting started guide](#) is just a click away.

Options for adding data sources to Metabase, including popular databases like MySQL, PostgreSQL, SQL Server, Amazon Redshift, BigQuery, and Snowflake.

# Usage Data Preferences



Your language is set to English



Hi, Mohamad Ikhsan . Nice to meet you!



I'll do self-service analytics for my own company



I'll add my own data later

5

## Usage data preferences

In order to help us improve Metabase, we'd like to collect certain data about product usage. [Here's a full list of what we track and why.](#)



Allow Metabase to anonymously collect usage events

Finish

If you feel stuck, [our getting started guide](#) is just a click away.

# You're All Set Up



I'll do self-service analytics for my own company



I'll add my own data later



We won't collect any usage events

## You're all set up!

✉ METABASE NEWSLETTER

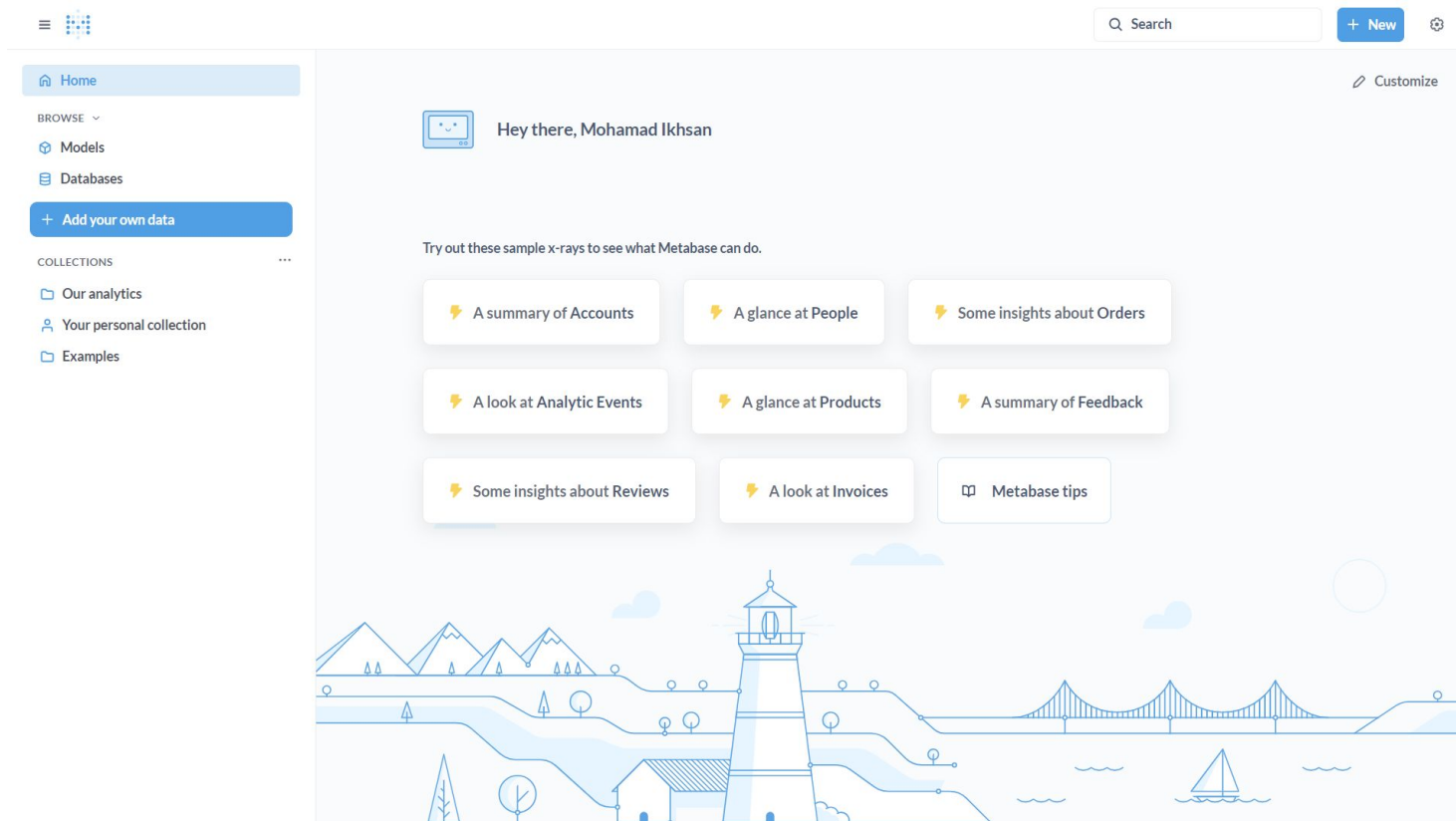
Get infrequent emails about new releases and feature updates.

Subscribe

Take me to Metabase

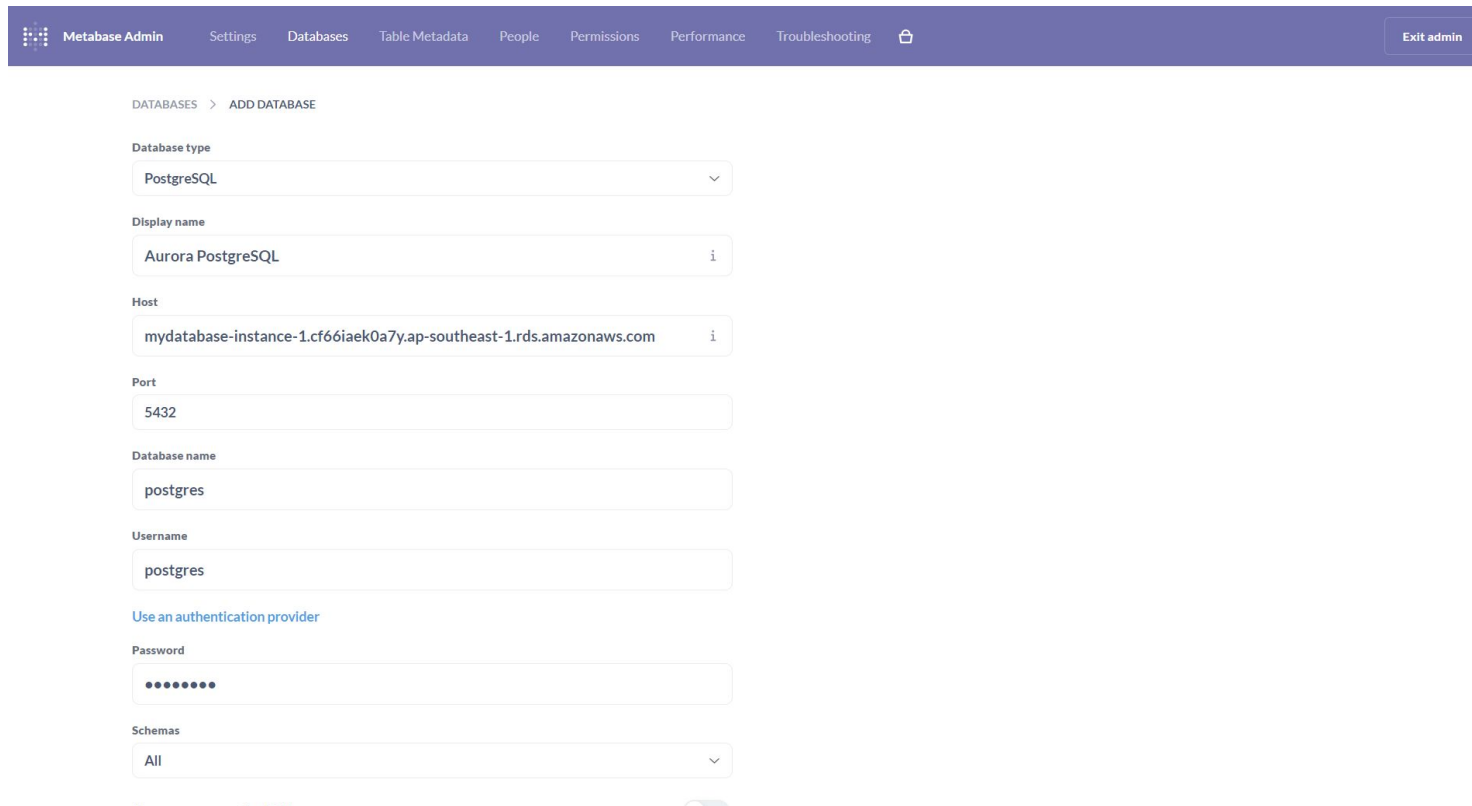
If you feel stuck, [our getting started guide](#) is just a click away.

# Metabase Home Page



Overview of the Metabase dashboard, including sample data exploration options and a summary of various analytics insights.

# Test Database Connection to Aurora PostgreSQL



The screenshot shows the Metabase Admin interface with a purple header bar. The header contains navigation links: Metabase Admin, Settings, Databases, Table Metadata, People, Permissions, Performance, Troubleshooting, and a lock icon. An 'Exit admin' button is in the top right. The main content area has a breadcrumb 'DATABASES > ADD DATABASE'. The form includes the following fields: 'Database type' (dropdown menu with 'PostgreSQL' selected), 'Display name' (text input with 'Aurora PostgreSQL'), 'Host' (text input with 'mydatabase-instance-1.cf66iaek0a7y.ap-southeast-1.rds.amazonaws.com'), 'Port' (text input with '5432'), 'Database name' (text input with 'postgres'), 'Username' (text input with 'postgres'), a link 'Use an authentication provider', 'Password' (password input with 8 dots), and 'Schemas' (dropdown menu with 'All' selected).

DATABASES > ADD DATABASE

Database type  
PostgreSQL

Display name  
Aurora PostgreSQL

Host  
mydatabase-instance-1.cf66iaek0a7y.ap-southeast-1.rds.amazonaws.com

Port  
5432

Database name  
postgres

Username  
postgres

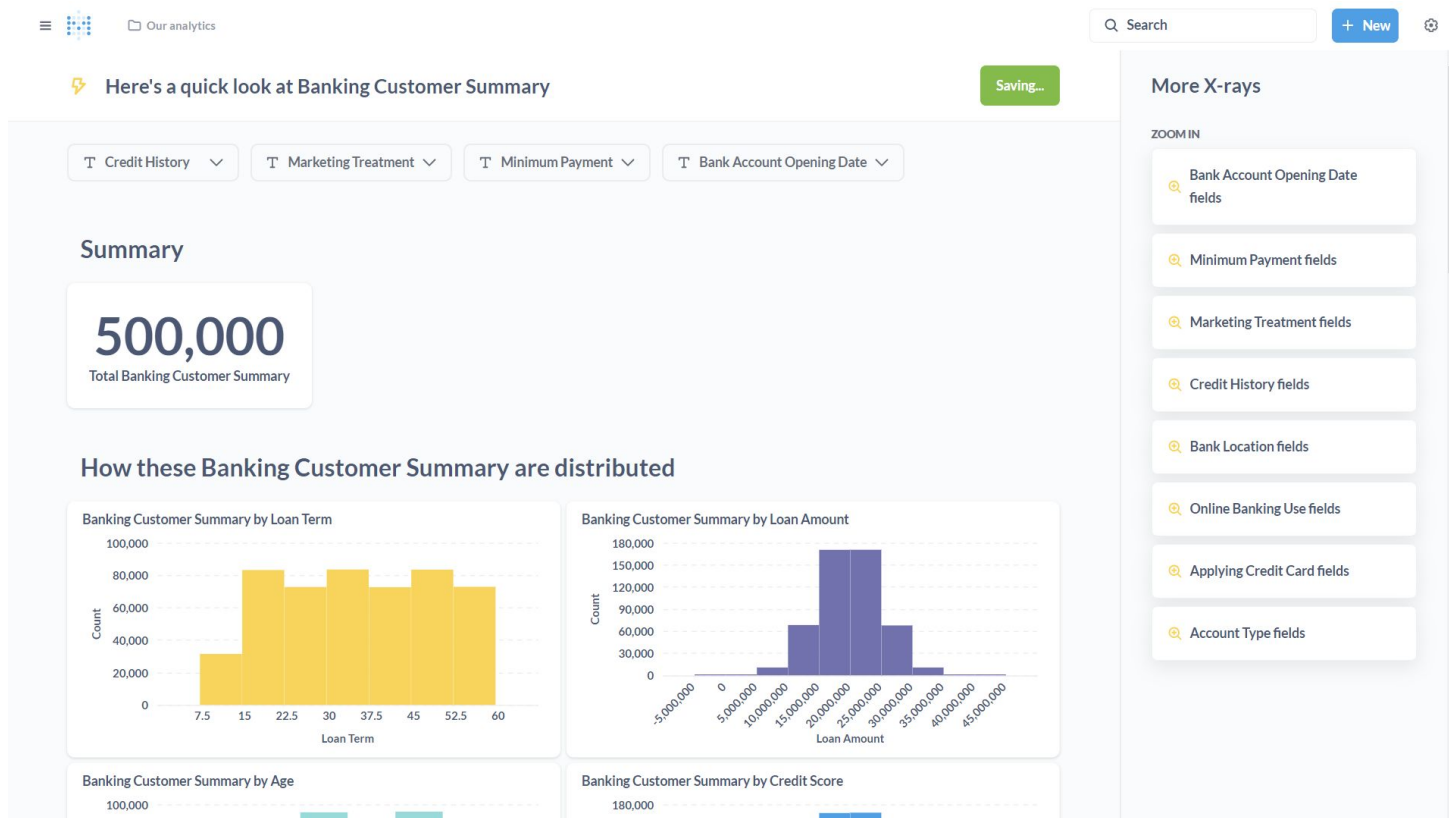
[Use an authentication provider](#)

Password  
••••••••

Schemas  
All

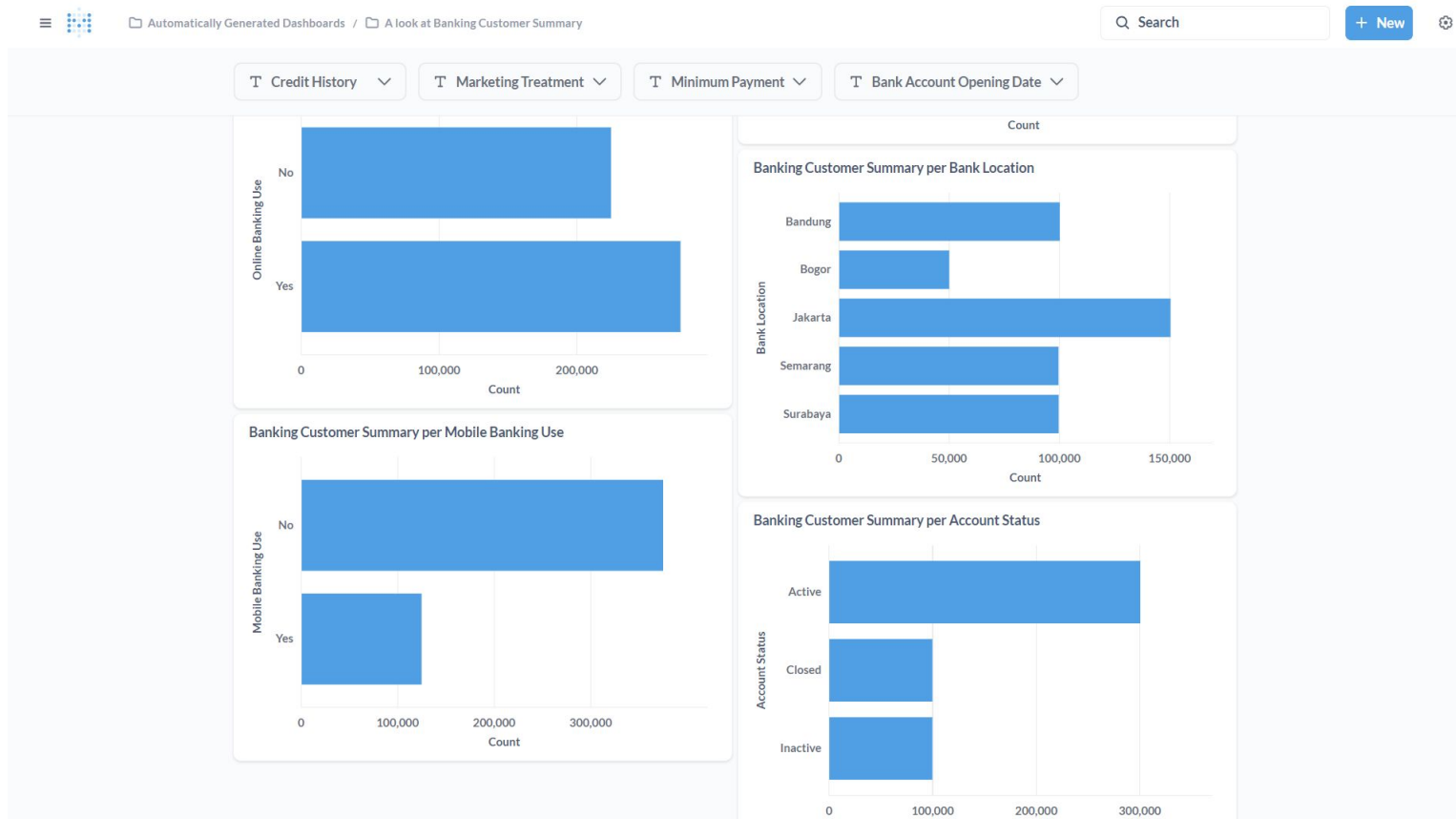
Steps to connect Metabase to a Aurora PostgreSQL database, including entering the host, port, database name, username, and password.

# Metabase Dashboard Using X-Ray



Metabase has an X-ray feature that generates a summary or visualization of a table or column to provide users with a quick understanding of their data.

# Metabase Dashboard Using X-Ray



A quick look at various analytics insights, such as banking customer summaries, loan terms, loan amounts, credit scores, and more.

# Conclusion and Recommendation

# Conclusion

- **Scalable Infrastructure:** Deploying Metabase on AWS EC2, combined with RDS Aurora PostgreSQL, provides a scalable and reliable data analytics platform.
- **Efficient Setup:** The step-by-step process ensures a smooth setup of both the database and Metabase, facilitating quick deployment.
- **Enhanced Data Visualization:** Users can leverage Metabase's capabilities for powerful data visualization and analytics.
- **AWS Integration Benefits:** The integration with AWS services ensures high availability, security, and performance for data-driven decision-making.

# Recommendations

- **Security:** Use AWS Secrets Manager, enable database encryption, and manage access with security groups and IAM roles.
- **Backups and Monitoring:** Implement regular RDS backups with automated snapshots and set up CloudWatch for monitoring and alerts.
- **Cost Optimization:** Review and optimize AWS usage, selecting the right instance types and using reserved instances or savings plans.
- **Updates and Training:** Keep Metabase and AWS services updated, and ensure thorough documentation and team training.

Thank You